Scikit-learn Quick Reference

1. Libraries used (the big picture)

Library	What it gives you	Import style
pandas	Load the CSV, groupby, cut, get_dummies	`import pandas as pd`
numpy	Numbers, arrays, log1p, cumsum	`import numpy as np`
matplotlib	Plots (scatter, elbow, dendrogram)	`import matplotlib.pyplot as plt`
scikit-learn (sklearn)	Models, scaling, metrics, CV, tuning, pipeline	`from sklearn.xxx import Yyy`
scipy	Dendrogram, random number distributions	`from scipy.xxx import yyy`
joblib	Save and load a trained model	`import joblib`

2. Topic 1 and 2: Unsupervised learning and clustering

What	Class / function	Where it is from	How to use it
K-Means	KMeans	sklearn.cluster	`KMeans(n_clusters=3, n_init=10).fit(X)`
Hierarchical clustering	AgglomerativeClustering	sklearn.cluster	`AgglomerativeClustering(n_clusters=3, linkage="ward")`
Merge tree maths	linkage	scipy.cluster.hierarchy	`Z = linkage(X, method="ward")`
Dendrogram plot	dendrogram	scipy.cluster.hierarchy	`dendrogram(Z)`
Group labels	.labels_	on the fitted KMeans	`model.labels_`
Centres	.cluster_centers_	on the fitted KMeans	`model.cluster_centers_`
Inertia (for elbow)	.inertia_	on the fitted KMeans	`model.inertia_`
Profile the groups	groupby().mean()	pandas	`df.groupby("cluster")[cols].mean()`
Silhouette score	silhouette_score	sklearn.metrics	`silhouette_score(X, labels)`
Davies-Bouldin	davies_bouldin_score	sklearn.metrics	`davies_bouldin_score(X, labels)`
Anomaly detection (named only)	IsolationForest	sklearn.ensemble	`IsolationForest().fit_predict(X)`

3. Topic 3: Dimensionality reduction

What	Class / function	Where it is from	How to use it
PCA	PCA	sklearn.decomposition	`PCA(n_components=2).fit_transform(X_scaled)`
Keep 95% variance	PCA(n_components=0.95)	sklearn.decomposition	`PCA(n_components=0.95).fit(X_scaled)`
Variance per component	.explained_variance_ratio_	on the fitted PCA	`pca.explained_variance_ratio_`
Running total	np.cumsum	numpy	`np.cumsum(pca.explained_variance_ratio_)`
How PCs mix columns	.components_	on the fitted PCA	`pca.components_`
t-SNE (named only)	TSNE	sklearn.manifold	`TSNE(n_components=2).fit_transform(X_scaled)`

4. Topic 4: Feature engineering

What	Class / function	Where it is from	How to use it
New column	plain pandas maths	pandas	`df["new"] = df["a"] / df["b"]`
Make bins	pd.cut	pandas	`pd.cut(df["age"], bins=[...], labels=[...])`
Polynomial / interaction	PolynomialFeatures	sklearn.preprocessing	`PolynomialFeatures(degree=2)`
Standard scaling	StandardScaler	sklearn.preprocessing	`StandardScaler().fit_transform(X)`

What	Class / function	Where it is from	How to use it
0 to 1 scaling	MinMaxScaler	sklearn.preprocessing	`MinMaxScaler().fit_transform(X)`
Log transform	np.log1p	numpy	`np.log1p(df["income"])`
Fill missing values	SimpleImputer	sklearn.impute	`SimpleImputer(strategy="median")`
Drop constant columns	VarianceThreshold	sklearn.feature_selection	`VarianceThreshold(threshold=0.0)`
Keep best k columns	SelectKBest, f_classif	sklearn.feature_selection	`SelectKBest(f_classif, k=10)`
Correlation check	.corr()	pandas	`X.corr()`
One-hot encoding	pd.get_dummies	pandas	`pd.get_dummies(df, columns=cat_cols)`
Different steps per column type (named only)	ColumnTransformer	sklearn.compose	`for number and text columns together`

5. Topic 5: Model evaluation (metrics)

What	Function	Where it is from	How to use it
Accuracy	accuracy_score	sklearn.metrics	`accuracy_score(y_test, y_pred)`
Precision	precision_score	sklearn.metrics	`precision_score(y_test, y_pred)`
Recall	recall_score	sklearn.metrics	`recall_score(y_test, y_pred)`
F1 score	f1_score	sklearn.metrics	`f1_score(y_test, y_pred)`
Confusion matrix	confusion_matrix	sklearn.metrics	`confusion_matrix(y_test, y_pred)`
Draw the matrix	ConfusionMatrixDisplay	sklearn.metrics	`ConfusionMatrixDisplay(cm).plot()`
All metrics in one print	classification_report	sklearn.metrics	`classification_report(y_test, y_pred)`

6. Topic 6: Cross validation (all from sklearn.model_selection)

What	Class / function	How to use it	Easy meaning
Train/test split	`train_test_split`	`train_test_split(X, y, test_size=0.2, stratify=y, random_state=42)`	One simple split
K-Fold	`KFold`	`KFold(n_splits=5, shuffle=True, random_state=42)`	K pieces, each takes a turn as test
Stratified K-Fold	`StratifiedKFold`	`StratifiedKFold(n_splits=5, shuffle=True, random_state=42)`	Same, keeps the class ratio equal
Repeated K-Fold	`RepeatedStratifiedKFold`	`RepeatedStratifiedKFold(n_splits=5, n_repeats=3)`	Whole K-Fold repeated, steadier score
Quick CV scores	`cross_val_score`	`cross_val_score(model, X, y, cv=cv, scoring="f1")`	One score per fold
CV with train score too	`cross_validate`	`cross_validate(model, X, y, cv=cv, return_train_score=True)`	Shows overfitting gap
CV predictions (named only)	`cross_val_predict`	`cross_val_predict(model, X, y, cv=cv)`	Prediction for every row
Leave-one-out, time series (named only)	`LeaveOneOut, TimeSeriesSplit`	`same module`	Special CV types

7. Topic 7: Hyperparameter tuning

What	Class / function	Where it is from	How to use it
Try every combination	GridSearchCV	sklearn.model_selection	`GridSearchCV(model, param_grid, cv=5, scoring="f1")`
Try random combinations	RandomizedSearchCV	sklearn.model_selection	`RandomizedSearchCV(model, param_dist, n_iter=15, cv=5)`
Random value ranges	randint	scipy.stats	`{"max_depth": randint(2, 20)}`
Best settings	.best_params_	on the fitted search	`grid.best_params_`
Best CV score	.best_score_	on the fitted search	`grid.best_score_`
Best trained model	.best_estimator_	on the fitted search	`grid.best_estimator_`
See a model's settings	.get_params()	any sklearn model	`model.get_params()`
Setting names in a pipeline	step__param	naming rule	`"model__max_depth"`

8. Topic 8: Pipeline

What	Class / function	Where it is from	How to use it
Chain steps (named)	Pipeline	sklearn.pipeline	`Pipeline([("scale", StandardScaler()), ("model", LogisticRegression())])`
Chain steps (short)	make_pipeline	sklearn.pipeline	`make_pipeline(StandardScaler(), LogisticRegression())`
Train everything	.fit()	on the pipeline	`pipe.fit(X_train, y_train)`
Predict	.predict()	on the pipeline	`pipe.predict(X_test)`
Save the trained pipeline	joblib.dump	joblib	`joblib.dump(pipe, "best_pipeline.joblib")`
Load it back	joblib.load	joblib	`joblib.load("best_pipeline.joblib")`

9. The models used in the examples

Model	Class	Where it is from
Logistic regression	LogisticRegression	`sklearn.linear_model`
Linear regression (your example)	LinearRegression	`sklearn.linear_model`
Decision tree	DecisionTreeClassifier	`sklearn.tree`
Random forest	RandomForestClassifier	`sklearn.ensemble`
KNN	KNeighborsClassifier	`sklearn.neighbors`
Practice dataset	load_breast_cancer	`sklearn.datasets`

10. The 3 methods that appear everywhere

Method	What it does	Used on
`.fit(X, y)`	Learn from data	models, scalers, PCA, pipelines
`.transform(X)`	Apply what was learned	scalers, PCA, imputers
`.predict(X)`	Give answers for new data	models, pipelines